

# KAMALESHWARAN P

Mechanical Engineer · Design Specialist · FEA & Simulation · AI-Integrated Systems

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Coimbatore, Tamil Nadu

## PERSONAL INFO

**DOB:** 17 February 2007

**Blood Group:** B+ (ve)

**Nationality:** Indian

**Location:** Coimbatore, Tamil Nadu

**Languages:** English (Cambridge B2 Upper-Intermediate), Tamil (Native)

## EDUCATION

### B.E. Mechanical Engineering

Sri Ramakrishna Institute of Technology  
CGPA: 7.93/10 | 2nd Year (Ongoing) | Expected 2028

### 12th HSC — 80.1%

Kongunadu Matric Hr. Sec. School, Namakkal, 2023

### 10th SSLC — 87.6%

Kongunadu Matric Hr. Sec. School, Namakkal, 2021

## TECHNICAL SKILLS

### CAD / Design

AutoCAD · SolidWorks · Fusion 360 · Solid Edge

### Simulation / FEA

ANSYS Workbench · Fluent · Thermal · CFD

### Engineering Tools

MATLAB · Simulink · GD&T · 3D Printing

### Embedded Systems

Arduino · ESP32/ESP8266 · UART/I2C/SPI

### AI / Software

OpenCV · TensorFlow · SLAM · Neural Networks

### Programming

C · C++ · Embedded C · HTML5 · CSS3 · JavaScript

### Dev Tools

Git/GitHub · VS Code · Arduino IDE · Canva

## LEADERSHIP

### Design Head

ASME SRIT Student Section, 2023–Present

### Kart Team Lead & Design Head

SRIT Kart Team — TN Karting S02, Title Won

### Analysis Head

Kart Team — ANSYS FEA & Structural Analysis

### Lead Member

SAE India, SRIT Chapter, 2023–Present

## CAREER OBJECTIVE

Mechanical Engineering student at Sri Ramakrishna Institute of Technology (CGPA: 7.93/10) with advanced proficiency in CAD, FEA, and product development. Passionate about design engineering and building intelligent systems at the intersection of mechanical precision and technology. Cross-disciplinary skills in embedded systems, AI, and web development make me a versatile, future-ready engineer. Seeking a senior-level role in design, analysis, or product development, with openness to contribute wherever impact is greatest.

## INTERNSHIP EXPERIENCE

### Mechanical Engineering Intern — Sri Jayasurya Engineering, Perur, Coimbatore May–Jun 2026 (2 Wks)

- Contributed to robotic arm design and assembly — kinematic linkage design and fabrication workflows.
- Performed quality inspection: dimensional tolerancing, component verification and instrumentation.
- Strengthened competencies in manufacturing processes and systematic engineering problem-solving.

### Web Development Intern — ZyvOne Technologies, Udumalpettai, Tiruppur Jun 2026 (2 Wks)

- Engineered a hospital management website with appointment scheduling and a responsive patient-facing UI.
- Developed a jewellery e-commerce platform with product catalogue and mobile-responsive design.
- Gained exposure to real-world software delivery pipelines and agile project execution practices.

### Virtual Site Development Intern — Sri Ramakrishna Institute of Technology 2024 (3rd Sem Gap)

- Designed a comprehensive virtual campus platform with interactive mapping and multimedia integration.
- Led end-to-end project execution: visualisation, campus mapping, and digital outreach architecture.

## KEY PROJECTS

### Kart Design & Analysis — TN Karting Championship S02 — Kart Team Lead + Design Head + Analysis Head | Title Won

- Full 3D CAD modelling, chassis geometry, suspension layout and competition compliance verification.
- Conducted ANSYS FEA on chassis and critical components — data-driven iterative design leading to title win.
- Led team coordination, fabrication oversight and competition preparation end-to-end.

### Autonomous Rover — Caterpillar Autonomy Challenge — Shastra, IIT Madras Jan–Mar 2026 — Semi-Finalist

- Engineered full autonomous navigation using SLAM and monocular vision for real-time obstacle detection.

CERTIFICATIONS (cont'd)

- ◆ Event Management Professional
- ◆ Sales & Marketing Strategies
- ◆ Learnathon — SRIT

MEMBERSHIPS

|                |                                |
|----------------|--------------------------------|
| ASME           | Design Head, SRIT Chapter      |
| SAE India      | Lead Member, SRIT Chapter      |
| Rotaract Intl. | Int'l Service Director, SRIT   |
| IIC / HIVE     | Member & Innovation Ambassador |

Patent 03 — Published — India, 2025

- AI-Based Quality Inspection System for Industrial Quality Departments.

RESEARCH PAPERS

- "Performance Evaluation of Shell & Tube Heat Exchanger with Parabolic Solar Collector" — SRM Institute of Technology, National Conference (Published, Peer-Reviewed).
- "Energy Storage & CFD Analysis of Pebble Bed Heat Exchanger" — Dr. N.G.P Institute of Technology, National Conference.

PROFESSIONAL MEMBERSHIPS

| Organization   | Role                           | Duration     |
|----------------|--------------------------------|--------------|
| ASME           | Design Head, SRIT Chapter      | 2023–Present |
| SAE India      | Lead Member, SRIT Chapter      | 2023–Present |
| Rotaract Intl. | Int'l Service Director, SRIT   | 2023–Present |
| IIC / HIVE     | Member & Innovation Ambassador | 2023–Present |

REFERENCES

|                                                                                                                                                         |                                                                                                                                              |
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| <b>Mr. Immanuel R</b><br>Professor, Dept. of Mechanical Engineering<br>Sri Ramakrishna Institute of Technology<br>immanuel.me@srit.org   +91 9677817992 | <b>Ms. Sanjana N</b><br>Professor, Dept. of Computer Science & Engg.<br>Sri Ramakrishna Institute of Technology<br>sanjana.cse@sritcbe.ac.in |
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All information accurate as of June 2026 | [linkedin.com/in/kamaleshwaran-p](#) | [github.com/Kamaleshwaranbadboy](#)